Classical mechanics } · Know precise knowledge of positions, velocities, momentum
· Model the future evolution of system

Macro thermo } $N \sim 10^{23}$ particles
$6N$ degrees of freedom

We usually have information about the "macroscopic state" of system (V, E, T, etc.)
↳ How do we get the micro-state i.e. particular values of all $6N$ dof's.

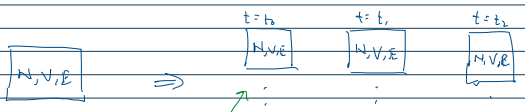
↓

Construct an ensemble of system → average behavior of a collection of identical N particle system distributed over a range of different possible states

They are not smaller systems or control volume

↓

More so representing the entire system & its time evolution w/ identical systems at different times (or initial conditions)



$\Omega = \sum$ ensembles (total ensembles)

TLDR:

- Each ensemble can be represented by a single point in $6N$ dimensional space ($3N$ position & $3N$ momentum)
- These ensembles are clubbed together in a phase space density ?

  Thus this cloud/collection of ensembles can be modelled like an incompressible flow

- This way we can use ensemble averages since if you wait long enough, a single system will visit every part of the phase space allowed by its energy

  ↓

  In a way the time evolution is just modelled as taking place already to give all the possible states → then you are just sampling from it.

  (This is the ergodic hypothesis → Boltzmann 1871)

Each ensemble has N particles whose dynamic behavior is given by Hamilton's eqn's of motion

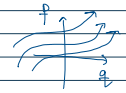
These are a set of first-order differential equations that describe how a system evolves through phase space (q_i, p_i)

H = total energy of system

$$\dot{q}_i = \frac{\partial H_i}{\partial p_i} \qquad \dot{p}_i = -\frac{\partial H_i}{\partial q_i}$$

↓

A point in phase space represents the complete state of system (position & momentum) → as time progresses this point traces a path called a trajectory



$f(p, q, t)$ = density in phase space

$\Omega = \int_{E \le \mathcal{H} \le E+\delta E} dp\,dq$ → volume of phase space whose energy lies b/w E and $E + \delta E$

↳ just a pure number quantum mechanically → no. of microstates for a given E

$$S = -k \sum_\nu P_\nu \ln(P_\nu)$$

(N, V, E) → constant

Micro canonical Ensemble

+ Describes an isolated system (no exchange w/ environment)

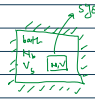
+ Principle of equal a-priori probability → every accessible microstate Ω is equally likely to be occupied

$$P_i = \begin{cases} \frac{1}{\Omega} & E_i = E \\ 0 & \text{else} \end{cases}$$

$$S = k_B \ln \Omega(N, V, E)$$

(energy can change)
(N, V, T) → constant

Canonical Ensemble



+ All microstates are not equally likely (since E is not constant) Probability of finding system in a microstate i w/ energy E_i given by the Boltzmann distribution

} specifically lower energies are favored $E_\nu \downarrow \rightarrow E_b \uparrow$

+ Probability of finding system in any microstate (ν) :

$$P_\nu = \frac{\Omega_b(E_b)}{\Omega(E)}, \quad E_b - E_\nu = E$$

↓ math & Taylor expansion

$$P_\nu = \frac{e^{-\beta E_\nu}}{\sum_\nu e^{-\beta E_\nu}}$$

} This is partition function (Q or Z)
+ Sum of the probabilities of all possible microstates equals 1
+ Has information about all the system's energy levels

} This is later used to bridge between micro and macro

$$A = -kT \ln Q$$

↓

For non-interacting independent particles the hamiltonian and hence the Energy microstates are separable

In the thermodynamic limit, both ensembles become equivalent

$$A = -kT \ln Q$$

$$U = \langle E \rangle = \left. \frac{-\partial \ln Q}{\partial \beta} \right|_{N,V} = \left. kT^2 \frac{\partial \ln Q}{\partial T} \right|_{N,V}$$

$$P = \left. k_B T \frac{\partial \ln Z}{\partial V} \right|_{N,T}$$

$$S = \frac{U - A}{T} = \left. k_B \ln Q + k_B T \frac{\partial \ln Q}{\partial T} \right|_{N,V}$$

For non-interacting, independent particles the Hamiltonian and hence the Energy microstates are separable

↓

Therefore we can assign

$$Q = \sum_\nu e^{-\beta E_\nu} = q_a q_b q_c \cdots$$

where $q_i = \sum_{\nu_i} e^{-\beta \epsilon_{\nu_i}}$ → molecular partition function

$q_i = q_i(V,T)$

Ideal gas

$$H = H_{trans} + H_{elec} + H_{vib}$$

quantum correction

$$Q = \frac{1}{h^{3N} N!} \int e^{-\beta H} \, dp \, dq$$